Gregory Polyakov

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Education

PhD **University of Tübingen & Max Planck Research School**, Computer Science — Tübingen, Germany — Oct 2024 – Aug 2028

- Health-NLP Lab, Supervisor: Prof. Dr. Carsten Eickhoff.
- Scholar, International Max Planck Research School for Intelligent Systems (IMPRS-IS).
- Research focus: Mechanistic interpretability of language models for NLP and IR tasks.
- Expected graduation: Aug 2028.
- Available for an internship in 2027 (flexible dates and duration).

MSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia — Sept 2021 – July 2024

- Research topics: sparse retrieval, multimodal retrieval, tool calling in LLMs.
- Thesis: "Approaches to Training Language Models for External Function Calling: Using Self-Generated Data for Better Error Detection and Recovery".

BSc **Moscow Institute of Physics and Technology**, Applied Mathematics and Computer Science — Moscow, Russia — Sept 2017 – July 2021

- Thesis: "Usage of MCMC Methods in Deep Learning".

Publications

Predicting Hallucinations Before, During, and After Generation with Sparse Autoencoder Features — 2026

Systematically studied hallucination detection from model internals before, during, and after generation, showing that a substantial signal exists before generation. Proposed PreSAGE, a trainable SAE-based detector with gated feature selection and attention pooling, best in 19 of 24 QA settings and the most consistent overall.
G. Polyakov, A. Bahrainian, C. Eickhoff
(Under review)

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance — 2026

Introduced the Wackiness Score and W-AUC to quantify semantically unrelated expansion terms across models with different vocabularies and sparsity levels. Trained 6 SPLADE variants, showing that the prevalence of wacky tokens increases with vocabulary size and that they mainly aid in-domain retrieval.
G. Polyakov, H. Scells, C. Eickhoff
arxiv.org/abs/2605.19628 (SIGIR 2026 (Reproducibility))

Interpretability Analysis of Arithmetic In-Context Learning in Large Language Models — 2025

Traced how 4 LLMs (up to 12B) use in-context examples for multi-step arithmetic, showing that format consistency matters more than arithmetic correctness. Distilled in-context examples into a function vector that raises zero-shot accuracy by up to 25 points and fully recovers one-shot accuracy on Pythia-12B.
G. Polyakov, C. Hepting, C. Eickhoff, A. Bahrainian
aclanthology.org/2025.emnlp-main.92 (EMNLP 2025 (Main))

Towards Best Practices of Axiomatic Activation Patching in Information Retrieval 2025
Proposed a noise-mitigating evaluation metric for activation patching in neural rankers that filters out spurious low-confidence signals. Systematically analyzed diagnostic datasets, identifying key selection criteria for queries and documents to ensure robust interpretability analysis.
G. Polyakov, C. Chen, C. Eickhoff
dl.acm.org/doi/10.1145/3726302.3730256 (SIGIR 2025 (Short))

ToolReflection: Improving Large Language Models for Real-World API Calls with Self-Generated Data 2025
Proposed a fine-tuning strategy that enables LLMs to self-correct API calls by learning from runtime execution feedback. Co-developed evaluation datasets for tool-augmented LLMs in realistic scenarios.
G. Polyakov, I. Alimova, D. Abulkhanov, I. Sedykh, A. Bout, S. Nikolenko, I. Piontkovskaya
aclanthology.org/2025.realm-1.14 (ACL 2025 (REALM Workshop))

Sinkhorn Transformations for Single-Query Postprocessing in Text-Video Retrieval 2023
K. Yakovlev, G. Polyakov, I. Alimova, A. V. Podolskiy, A. Bout, S. I. Nikolenko, I. Piontkovskaya
arxiv.org/abs/2311.08143 (SIGIR 2023 (Short))

Experience

Huawei Noah's Ark Lab, NLP Research Engineer (full-time) Moscow, Russia
Nov 2022 – Sept 2024
1 year 11 months
- Developed a fine-tuning method that teaches LLMs to detect and recover from failed API calls (ACL 2025 REALM Workshop).
- Designed and co-implemented the end-to-end NLU pipeline for a production in-car voice assistant in Russian.
- Co-developed a Sinkhorn-based post-processing method that improves text-video retrieval quality (SIGIR 2023).

Huawei Noah's Ark Lab, NLP Research Intern (full-time) Moscow, Russia
Oct 2021 – Nov 2022
1 year 2 months
- Co-developed the reranking model for a large-scale Russian-language product search engine: improved ranking quality by 5% relative NDCG@10 and cut inference latency 1.5x by adapting ColBERTer and TinyBERT-style distillation.

Invited Talks

Understanding Wacky Weights: A Dissection of SPLADE's Learned Term Importance Glasgow IR Seminar, University of Glasgow
Oct 2026

Scholarships

Presidential Grant Scholarship Feb 2018
Feb 2018 – Aug 2021. Four-year stipend for academic and competition merit, one of 1,200 recipients nationwide.

Teaching

Teaching Assistant: Introduction to AI University of Tübingen
Winter Semester 2026/27. Conducted hands-on tutorials and evaluated student assignments.

Teaching Assistant: Modern Search Engines University of Tübingen
Summer Semesters 2025 and 2026. Conducted hands-on tutorials, evaluated assignments, and mentored student projects.

Teaching Assistant: Introduction to Python Programming University of Tübingen
Winter Semesters 2024/25 and 2025/26. Conducted hands-on tutorials and evaluated student assignments.

Skills

Programming: Python, C/C++, Bash

ML & NLP: PyTorch, Hugging Face Transformers, PEFT, LLM fine-tuning (SFT, LoRA), tool use, Scikit-learn

Interpretability: TransformerLens, activation patching, steering, probing, SAEs, cross-layer transcoders

IR: Pyserini, FAISS, sparse and dense retrieval, neural reranking, knowledge distillation

Infrastructure: SLURM, multi-GPU training (HF Accelerate), Git

Languages

Russian: Native

English: C1 (IELTS Band Score 7)